

Name: Thomas, Edward, Gadiel, Robert

Date: 2/1/24

Class: Engineering Systems

MyDesign Portfolio

Element A Initial Template

How to use this template

Students: The details below are commonly found in problem definitions, but the specific assignment given by your instructor may dictate different details. If you make a copy of this template to edit for your own problem definition, be sure to delete the instructions.

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Explain your problem in detail, including details that you might consider common knowledge. Use data and statistics to justify your problem statement. Do not provide a bulleted list of facts; rather, organize the problem details in paragraph form. Address as many of the 5 Ws +1 H (who, what, where, when, why, how) as is possible, for example:

- 1) Who is involved in this problem?
- 2) What are they doing?
- 3) What unique situations does the problem occur within, if any?
- 4) What is the size of the issue?
- 5) Where is the problem happening?
- 6) When does the problem occur?
- 7) Why does the problem happen?
- 8) How often does the problem happen?

In order to earn a score of 5, you will need to **cite at least five sources** that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. **Use in-line citations to explicitly state which information is from which source.** Do not just list all of the citations at the end of the submission or in footnotes.

Stakeholder Concerns and Connections (A3, A4)

Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. **Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem.** Consider the breadth of the problem and how it might have social and

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environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.

Problem Importance (A6)

End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.

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1. Who is involved in this problem?

- a. Mr L.
- b. Students
 - i. Mr. L's students
 - ii. Robotics Club
 - iii. Previous students of Mr. L
- c. Custodians
- d. Machinery manufacturers (Not directly affected)

2. What are they doing?

- a. Mr. L
 - i. Teacher and supreme overseer of the Design Room and Workshop. He is in charge of taking care of the machinery and the classroom, and (attempting to) keep the area clean and free of dust.
 - ii. Sometimes he uses the shop for himself, and he sets up the current ventilation system, and as the primary caretaker he pays more attention to the room's organization and upkeep than students
- b. The students
 - i. Primary customers of the woodshop, operating the machinery to build assigned and personal projects alike, use machinery like the band saw or the sander, producing a large amount of sawdust
 - ii. Less conscious of their actions and impact on the room
- c. The custodians
 - i. While not directly responsible for taking care of the woodshop, custodians still clean the classroom outside, and an influx of dust piling in the corners is just more mess for them to clean up
- d. The manufacturers are the people making the machines which produce the dust
 - i. While not directly involved in this issue, dissatisfaction with the current built in methods of containing dust could deter future customers from buying their products

3. What unique situations does the problem occur within, if any?

- a. Specific to this issue, students often face the issue of having sawdust fly off of the machinery and coating their face and hair, leaving them tan and dirty, triggering asthma attacks and polluting clothing.
- b. The sawdust produced in the workshop also impacts the environment, tracking put through windows and on students

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4. What is the size of the issue?

- a. Local, the issue exclusively originates from the woodshop, as it is where the machinery and woodworking tools are located. However, some sawdust is able to leak out into the classroom and courtyard outside, but that is its furthest extent. A moderate amount of dust is produced when someone uses the machinery, especially noticeable during operation of the sander.

5. Where is the problem happening?

- a. The issue occurs exclusively in the woodshop, but is able to leak into the classroom.

6. When does the problem occur?

- a. The problem itself, the spread of sawdust, happens whenever a student or person uses the equipment to produce dust, then tracks it outside of the workshop by leaving the door open, misoperation of the ventilation, or just tracking it out with their clothes and shoes.

7. Why does the problem happen?

- a. The issue happens because people lack awareness of the issue of dust, and that the current solutions to the dust (ventilation and vacuums) are not able to keep up with the production of dust, causing a buildup.

8. How often does the problem happen?

- a. The problem happens fairly consistently, as most of them happen unintentionally or without knowledge. Students and Mr. L use the woodshop quite a lot, as projects are fun to make, and the robotics team often needs new parts and materials to be made to adapt to new situations.

Primary Stakeholders:

1. Mr. L
 - a. As the teacher of the classroom and primary caretaker of the workshop, Mr. L holds the most responsibility and “stake” in this issue, as damages caused by the dust can cause issues in His tools, His technology, and His students.
2. Mr. L’s Current Students
 - a. As the Students of Mr. L, this group is composed of kids at Marshall who attend his class, build projects in the workshop, and inhabit his classroom. If Mr. L’s class is messy and unkempt, then his students will be dissatisfied with the experience, and with Mr. L as a teacher, making them less likely to take his class in the future. Any dust that escapes through the windows of the

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workshop may make their way to the courtyards of the school, polluting the environment and lungs of students.

3. Mr. L's Previous Students who use the shop
 - a. Any students who had Mr. L as a teacher in the past, and continue to use the workshop either for clubs or personal use may find the current state of dust in the workshop to be unsatisfying, and will be further dissuaded from taking his classes further
4. Robotics Team
 - a. Gryphon Robotics also use the workshop for construction of parts and materials, so if the workshop is untidy then they will be less efficient, and less likely to do well at competitions, causing losses both for the team and for the member's reputations
5. Custodians
 - a. While not inherently in charge of cleaning the workshop, any attempts to clean the classroom attached to it, or the surrounding halls, may be met with excess dust that would annoy or frustrate them.

Conclusion:

The amount of dust prevalent within the workshop and classroom of Mr. L poses a major threat to the health and safety of students and faculty alike. The Department of Health notes that the presence of sawdust heavily triggers respiratory issues like asthma or bronchitis, conditions that students may have, causing damage and despair.

Work Cited:

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